KHIZAR ANJUM

Machine Learning Engineer

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154 citations, h-index: 7, i10-index: 7

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, CPS Lab, Rutgers University — (2019 – Present) | Piscataway, NJ, US

- Specialized in developing efficient deep learning architectures for various applications, with focus on **resource-constrained devices**. Published **4 journal articles** and **14 conference papers** in prestigious venues.
- Filed **3 patents** and helped secure funding worth **$1.135 million** in prestigious NSF awards.
- Invented **ultra-low power circuits** for implementing neural network with only micro-watts of power-consumption.
- Developed **ultra-light crowd pattern identification** methods suitable for on-board computation.
- Collected first-ever underwater **acoustic communications dataset** from NJ for channel estimation.

Research Assistant, Communications Lab, LUMS — (2017 – 2019) | Lahore, Pakistan

- Developed and trained cutting-edge **Convolutional Neural Network classifier** with **Bi-LSTM evidence aggregation** model for Parkinson's detection on **multimodal data**.
- Achieved **89% Area under ROC Curve performance** compared to baseline of 56%, a **23% improvement**.

RELEVANT SKILLS

- **Programming & Languages:** Python (advanced), C++, JavaScript, SQL, MATLAB, Bash, Chisel HDL
- **ML/AI & Generative AI:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, prompt engineering, instruction tuning, RAG, foundational model fine-tuning
- **MLOps & Deployment:** Docker, CI/CD, FastAPI, Flask, model monitoring, cloud-based ML deployment
- **Cloud & HPC:** AWS, Azure (Databricks, Azure AI Studio), GCP (Vertex AI), SLURM, distributed training
- **Data Engineering:** Data pipelines, ETL processes, SQL/NoSQL databases, Pandas, data preprocessing, Tableau
- **Development Tools:** Git/GitHub, Linux, VS Code, Jupyter, version control, OpenCV, GNU Radio, ROS

RESEARCH INTERESTS

Analog and Neuromorphic Computing, Ultra-Low Power Neural Networks, Crowd Monitoring, Embedded Machine Learning, Computer Vision, Signal Processing, Underwater Acoustic Communications, Autonomous UAVs, Hardware-Software Co-design, FPGA Acceleration

ACHIEVEMENTS & AWARDS

- Recipient of **Rutgers Innovation Awards** in the Graduate Student Category, 2025
- Recipient of **Paul Panayotatos Scholarship** for Sustainability-aware Electrical Engineering, 2024
- Selected for **NSF NOVUS & NSF I-Corps** program for innovation & customer discovery training, 2022
- Recognized with **Gold Medal** at LUMS for academic excellence, 2019
- Won **TA Achievement Award**, 2020 from Rutgers

LEADERSHIP EXPERIENCE

- Served as Program Coordinator for **two years** on ECE Department's **first-ever Graduate Student Association (GSA)**.
- Mentored **two** Capstone groups who got **first place** and **best-for-commercialization** awards at Rutgers University.

EDUCATION

Rutgers University, New Brunswick, NJ, (2019–2026 [Expected])

- PhD Candidate & Master of Science (MS), Electrical and Computer Engineering (ECE)
- Grade Point Average (GPA): 3.92/4; *Advised by* Dr. Dario Pompili

Lahore University of Management Sciences (LUMS), Lahore, Pakistan (2015–2019)

- Bachelor of Science (BS), Electrical Engineering (EE) – Gold Medalist
- Grade Point Average (GPA): 3.86/4; *Advised by* Dr. Muhammad Tahir and Dr. Momin Uppal

PATENTS

- **WO 2025/064997:** Anisotropic Diffusion-based Analog Neural Network Architecture
- **WO 2024/107672:** Techniques for Image Transmission through Acoustic Channels in Underwater Environments
- **(Pending Application):** Methods and Systems for Determining Group Motion Patterns

SELECTED RESEARCH PROJECTS

Ultra-Low Power Analog Neural Network Design for Health Monitoring — 2022-25

- Scaled existing neural network by writing a **custom Python-based generation script** for ngspice circuit simulator.
- Developed the idea of Folded Neural Network (FNN), thereby trading off on-chip space and power-consumption for duty-cycle and demonstrating that a **twice-folded 6-layer network** be operated at an 53µW.
- **Invented a novel analog architecture** called CvPU (Convolutional Processing Unit) that uses anisotropic diffusion to implement a convolutional layer with a **8-DoF 3 × 3 kernel** when tessellated, consuming only **55.58 mW**.
- Tested and implemented the digital-side neural networks on FPGAs such as TinyFPGA BX and Xilinx Ultra96-v2 by **writing custom neural computation modules** using Chisel hardware construction language.

Crowd Prediction and Behavior Assessment Using Adaptive UAVs — 2024-25

- **Leveraged efficient signal processing techniques** by piggy-backing inter-prediction results from MPEG-4 hybrid video-encoder for crowd pattern identification to deliver **45× reduction** in execution time compared to baseline.
- Delivered comparable performance to Deep Learning and other optical flow-based baselines for crowd pattern identification task with Average Angular Error (AAE) of **14.99° on TUB CrowdFlow Dataset video sequences**.
- Implemented **automated exponentially converging implementation** of Volkov Block Method for solving Laplace Equation, identifying crowd behavior based on domain. https://github.com/khizar-anjum/pyBlockMethod

Deep Joint Source Channel Coding for Underwater Image & Video Transmission — 2022-25

- Developed an **LSTM-based sequence-to-sequence model** to enable channel-state aware deep joint-source channel coding for transmitting images through underwater acoustic channel.
- Implemented a variable encoder that **outperforms traditional JPEG/JPEG2000 approaches** for compression in low-SNR regimes, **i.e., from 0 to 13 dB, and avoids cliff-effects** observed in brittle traditional techniques.
- Collected **first-ever real-life acoustic communication dataset** under various center frequency, bandwidth, and TX/RX configurations at Barnaget Bay, NJ and Atlantic City, NJ and **published the dataset as ACommSet**.

On-board Inference on Unmanned Aerial Vehicles — 2020-25

- Developed and implemented Convolutional Bi-LSTM auto-encoder for fault-cause detection on a PYNQ-Z1 FPGA yielding **40× speedup** on-board (runtime of 2.63 msec) with **a 96.25% test accuracy** for real-life experiments.
- Published a novel context-aware framework for efficient and robust segmentation on-board demonstrating **an mIOU of 0.95 on varied simulated weather conditions** such as sunny, foggy, and rainy.
- Developed a real-time video-based navigation framework for drones using adaptive parameter selection for object detection models, demonstrating **20-70% reduction in execution time** with only **2% of accuracy drop**.

SELECTED PERSONAL PROJECTS

Communication-aware Large Language Model Training on HPC Cluster — 2025

- Trained llama-3-8B and llama-3.3-70B-Instruct models using Megatron-LM on NERSC Perlmutter GPU cluster.
- Observed communication patterns for pipeline-parallelism, model-parallelism, and tensor-parallelism.

Model Context Protocol (MCP) Design for Security & Legal Applications — 2025

- Won 3rd prize in an open-hackathon for designing an MCP server for cybersecurity vulnerability assessment, providing tools to search GitHub repositories, query NIST NVD, access CISA's Known Exploited Vulnerabilities catalog, and analyze repository contents for CVE research. https://github.com/khizar-anjum/risky-business-mcp
- Designed a powerful MCP server for legal research across 3,352 U.S. courts using the CourtListener API, enabling access to millions of legal documents and PACER data. https://github.com/khizar-anjum/courtlistener-mcp